



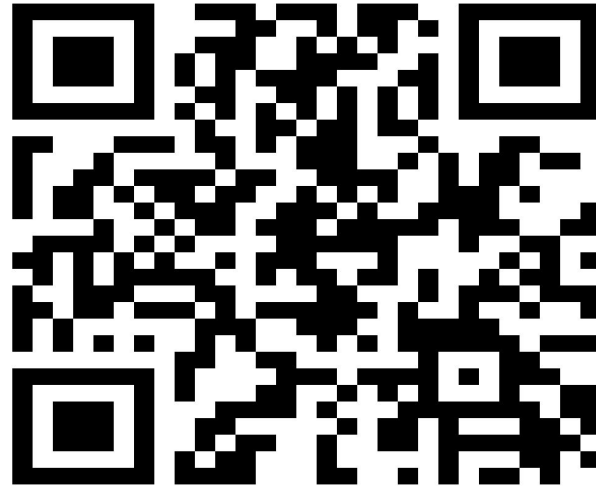
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TEST TIME ON VEDIC MATHS TECHNIQUES

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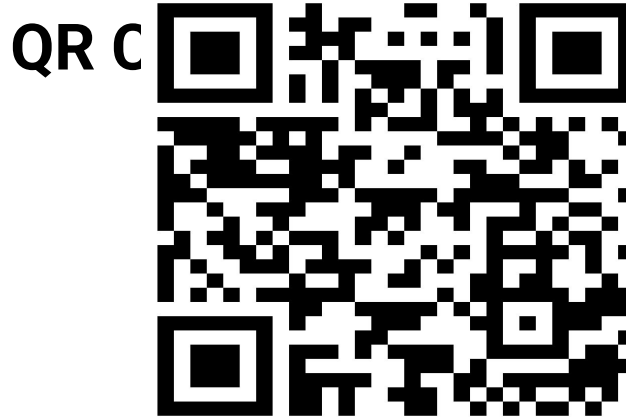


MULTIPLICATION SHORTCUTS



TEST TIME ON VEDIC MATHS

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MULTIPLICATION SHORTCUTS

1st Trick-Two conditions

- If the adjacent numbers are same
- If the unit digits are add up to 10.

Ex: $53 * 57 = ?$

$$(5*6)(3*7) = 3021$$

{Next digit of 5 is 6 is being multiplied with first term and unit term is multiplied}



Concepts

2nd Trick- Base value as 10's

Ex: $9 \times 8 = ?$

$$\begin{array}{r} 9 \quad -1 \\ * 8 \quad -2 \\ \hline \text{Sum } 7 \quad 2 \end{array}$$

$$; \text{sum} = (9 + (-2)) = 7 \text{ or } (8 + (-1)) = 7$$

$$; \text{product} = (-1 \times -2) = 2$$

Product ; Product term should be equal
to the number of zeros in base value

Ex: $98 \times 97 = ?$

$$\begin{array}{r} 98 \quad -2 \\ 97 \quad -3 \\ \hline \text{sum } 95 \quad 06 \text{ Product} \end{array}$$



Concepts

3rd Trick Base Deviation

Ex: $13 \times 8 = ?$

13	+3
8	-2
11	-6
-1	+10
10	4
10	4

4th Trick Multiplication in Base value

Ex: $61 \times 64 = ?$

61	+1	
	64	+4
	65	4
*6		
	390	4



Concepts

5th Trick

Ex: $33 * 21 = ?$

$$33 * 21 = 0603$$

$$\begin{array}{r} 09 \\ \hline 693 \end{array} (3*2)+(3*1)$$

6th Trick- when multiplied with 11 or 12

Ex: $23 * 11 = ?$

Step 1: write first term

step 2: add up with upcoming term

step 3 :write the last term

$$\begin{array}{r} 23 * 11 \\ \hline 253 \end{array}$$



Concepts

6th Trick- when multiplied with 11 or 12

Ex: $123 * 12 = ?$

Steps: First write the first term and multiply the first digit * 2 and add the next digit and follow the same

$$\begin{array}{r} \text{---} 123 * 12 \\ \text{---} 1\ 4\ 7\ 6 \text{---} \end{array} \quad \{ 1; 1*2+2; 2*2+3; 3*2+0 \}$$



Concepts

7th trick-when multiplied by 5

Ex: $38 * 5 = ?$

$$38 * \frac{10}{2} \Rightarrow 38 / 2 * 10 = 190$$

5 can be written as $10/2$

2 is brought to the denominator of 38 and is done.

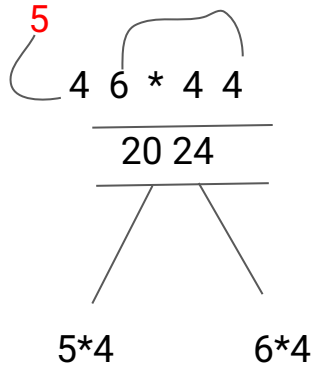
Question: 01

$46 \times 44 = ?$

- A. 2020
- B. 2024
- C. 2021
- D. 2012

Answer: B

Explanation:



Question: 02

$$31 * 39 = ?$$

- A. 1209
- B. 1309
- C. 1039
- D. 1299

Answer: A

Explanation:

$$\begin{array}{r}
 \overset{4}{\text{C}} \quad \text{3 1 * 3 9} \\
 \hline
 \text{12 09} \\
 \hline
 \text{4*3} \quad \text{9*1}
 \end{array}$$

The diagram illustrates a multiplication process. At the top, the number 31 is followed by an asterisk and the number 39. A red '4' is positioned above the first '3' of '31', with a curved line connecting it to the first '3' of '39'. A horizontal line separates this from the next row, which shows '12 09'. Another horizontal line follows. Below this, '4*3' is aligned under '12' and '9*1' is aligned under '09'. Diagonal lines connect '12' to '4*3' and '09' to '9*1'. A curved line also connects the '4' to the '3' in '4*3'.



Question: 03

$69 * 61 = ?$

- A. 5209
- B. 3208
- C. 4209
- D. 4005

Answer: C

Explanation:

$$\begin{array}{r}
 \overset{7}{\curvearrowright} 6 \quad 9 \quad * \quad 6 \quad 1 \\
 \hline
 42 \quad 09 \\
 \hline
 \swarrow \quad \searrow \\
 7*6 \quad 9*1
 \end{array}$$



Question: 04

$12 * 13 = ?$

- A. 166
- B. 155
- C. 156
- D. 163

Answer: C



Explanation:

Base value =10

	1 2	+2	
	* 1 3	+3	
Sum	15	6	product

$$12-10=2$$

$$13-10=3$$

$$\text{sum}=12+3 \text{ or } 13+3$$

$$\text{product}=2*3$$



Question: 05

$$15 * 19 = ?$$

- A. 285
- B. 280
- C. 396
- D. 425

Answer: A

Explanation:

Base value =10

	1	5	+5	
*	1	9	+9	
24	45			
	+4			
sum	28	5	product	

$$15-10=5$$

$$19-10=9$$

$$\text{sum}=15+9 \text{ or } 19+5$$

$$\text{product}=5*9$$



Question: 06

$102 * 104 = ?$

- A. 10604
- B. 12064
- C. 15608
- D. 10608

Answer: D

Explanation:

Base value 100

$$\begin{array}{r}
 102 \quad +2 \\
 * 104 \quad +4 \\
 \hline
 \text{sum} \quad 106 \quad | \quad 08 \quad \text{product}
 \end{array}$$

sum=104+2 or 102+4

product=2*4



Question: 07

$105 * 97 = ?$

- A. 10195
- B. 12056
- C. 10185
- D. 13185

Answer: C

Explanation:

Base value = 100

	1	0	5	+5	
*		9	7	-3	
	1	0	2	-15	
		-1		100	
sum	1	0	1	8	5 product

sum=105-3 or 97+5

product=5 * -3



Question: 08

$1007 * 998 = ?$

- A. 1004986
- B. 1005566
- C. 1002541
- D. 125864

Answer: A



Explanation:

Base value =1000

	1 0 0 7	+ 7	
	* 9 9 8	-2	
	1 0 0 5	-1 4	
	- 1	+1000	
sum	1 0 0 4	9 8 6	product

sum=1007-2 or 998+7

product=7*-2



Question: 09

$23 * 99 = ?$

- A. 2277
- B. 2266
- C. 3377
- D. 4477

Answer: A



Explanation:

$$\begin{array}{r}
 2 \ 3 \\
 * \quad 9 \ 9 \\
 \hline
 23-1 \quad 99 \\
 22 \quad -22 \\
 \hline
 22 \quad 77 \\
 \hline
 \end{array}$$



Question: 10

$43 * 17 = ?$

- A. 731
- B. 721
- C. 651
- D. 741

Answer: A



Explanation:

$$\begin{array}{r}
 4 \ 3 \ * \ 1 \ 7 \\
 \hline
 0 \ 4 \ 2 \ 1 \\
 3 \ 1 \\
 \hline
 7 \ 3 \ 1 \\
 \hline
 \end{array}$$



Question: 11

$31 * 24 = ?$

- A. 944
- B. 744
- C. 644
- D. 544

Answer: B



Explanation:

$$\begin{array}{r}
 \text{3 1 * 2 4} \\
 \hline
 \text{0 6 0 4} \\
 \text{1 4} \\
 \hline
 \text{7 4 4} \\
 \hline
 \end{array}$$



Question: 12

$34 * 11 = ?$

- A. 264
- B. 164
- C. 374
- D. 334

Answer: C



Explanation:

$$\begin{array}{r}
 + \\
 \overline{34 * 11} \\
 374
 \end{array}$$



Question: 13

$$68 * 5 = ?$$

- A. 340
- B. 240
- C. 140
- D. 440

Answer: A

Explanation:

$$68 * 5 \Rightarrow 68 * 10/2 \Rightarrow 68/2 * 10 \Rightarrow 24*10 = 240$$

Question: 14

$164 * 12 = ?$

- A. 1968
- B. 1966
- C. 1788
- D. 1668

Answer: A

Explanation:

$$\begin{array}{r}
 *2 \\
 \text{└─} 1\ 6\ 4\ * \ 1\ 2 \\
 \hline
 \text{①} \\
 1\ 8\ 6\ 8 \\
 \hline
 1\ 9\ 6\ 8 \\
 \hline
 \end{array}$$



Question: 15

198 * 12=?

- A. 2376
- B. 2366
- C. 2375
- D. 2333

Answer: A

Explanation:

$$\begin{array}{r} 198 * 12 \\ \hline 1212 \\ 1166 \\ \hline 2376 \end{array}$$





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THANK YOU